

Builder Basics — The Living Village (Extension)

Topic: Bring It All Together · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 90 minutes

🌱 This is a bonus challenge, not a weekly lesson. Try it after **Complete Your Village**, once you can build with loops, move the agent between spots, and place water and animals.

1 · The Challenge

You already finished a small village. Make one that feels **alive** — crops growing, animals wandering, water you can cross.

By the end, your village must have **all eight** of these:

 A **farm** — rows of crops  A **farm house** — where the farmer lives  A **house with a watchtower** — a home with a tall tower on top  A **fence** — around the farm or the animals  A **pond with fish** — water with at least one fish in it  A **bridge** — one that crosses over the pond A **well** — for water A **few animals** — cows, sheep, pigs, or chickens

There is no fixed size and no map. **You** decide how big each thing is and where it goes. One rule you must not break:

⚠️ **Everything you can build must be built with code.** No placing blocks by hand — every crop, fence post, and bridge plank goes down from your commands. (Animals are spawned — code or a spawn egg is fine.)

In your own words, list the eight things your village must have when you finish.

The last village was built with code. Why does building with code get *easier* the bigger your village gets?

2 · Plan Before You Build

A builder plans first. For **each** part, decide what it is made of and how big. Think about the blocks and whether a loop can do the repeating.

Farm — What crop? How many rows, and how long is each row?

Farm house — What blocks for the walls and roof? How wide and how tall?

House with a watchtower — Where does the tower sit — a corner, the middle, the roof? How much taller than the house is it?

Fence — What does it go around? How will the agent turn the corners?

Pond with fish — How will you make the dip in the ground? What fills it, and where do the fish go?

Bridge over the pond — How wide is the pond at its narrowest? How long must the bridge be to reach across?

Well — What shape is the hole? What goes around the top? Is there water inside?

Animals — Which animals, and how many? Will they go inside the fence?

3 · Think in Loops

Eight structures is a lot to place one block at a time. A loop is for anything you do **again and again** — a row of crops, the four sides of a fence, the planks of a bridge, the levels of a tower.

Pick three parts that will use a loop. For each one, say what the loop repeats each time.

Between one structure and the next, the agent has to **walk to a new spot** so nothing lands on top of the last build.

After you build the farm house, how does the agent get to a clean spot to start the watchtower house?

The bridge and the fence are special: they change direction. A fence goes across, then **turns**, then goes across again.

What has to happen to the agent between one side of the fence and the next?

4 · Show Your Work 📷 🎥

Now go to your homework world and build. Run your code for each part, one at a time, and move the agent between them. Build the water first, then put the bridge across it, so you can see it reach.

Keep going until your village has **all eight** parts — the farm, the farm house, the watchtower house, the fence, the pond with fish, the bridge, the well, and the animals.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.